

MTH603 MID TERM SOLVED MCQS
(VISIT VURANK FOR MORE)

Question # 1 of 10 (Start time: 04:40:08 PM)

Total Marks: 1

The determinant of a diagonal matrix is the product of the diagonal elements.

- 1. True
- 2. False

Question # 2 of 10 (Start time: 04:40:58 PM)

Total Marks: 1

Power method is applicable if the eigen vectors corresponding to eigen values are linearly independent.

- 1. True
- 2. false

A 3 x 3 identity matrix have three and different eigen values.

- 1. True
- 2. False

If $n \times n$ matrices A and B are similar, then they have the different eigenvalues (with the same multiplicities).

- 1. True
- 2. False

The Jacobi's method is a method of solving a matrix equation on a matrix that has ____ zeros along its main diagonal.

-
- 1. No
 - 2. At least one

An eigenvector V is said to be normalized if the coordinate of largest magnitude is equal to ____.

-
- 1. Unity
 - 2. zero
-

The Gauss-Seidel method is applicable to strictly diagonally dominant or symmetric positive definite matrices A.

1. True
2. False

The determinant of a _____ matrix is the product of the diagonal elements.

1. Diagonal
2. Upper triangular
3. Lower triangular
4. Scalar

Waisay main nay is ka answer **Diagnol** keea tha....par yeh charon options theek hain....
You can confirm it from internet...

Jab main yeh MCQ kar raha tha tou tab hi mujhay is par shak ho raha tha....kyun k upper aur lower triangular matrices tou linear algebra mein bhi bahut ziada bataye gaye tou yeh property wahan se hi yaad thi...

Eigenvalues of a symmetric matrix are all _____.

1. Real
2. Zero
3. Positive
4. Negative

The Power method can be used only to find the eigen value of A that is largest in absolute value—we call this eigen value the dominant eigen value of A.

1. True
2. False

The characteristics polynomial of a 3x 3 identity matrix is _____, if x is the eigen values of the given 3 x 3 identity matrix. where symbol ^ shows power.

1. $(x-1)^3$
2. $(x+1)^3$
3. x^3-1
4. x^3+1

For differences methods we require the set of values.

1. True
2. False

If $n \times n$ matrices A and B are similar, then they have the different eigenvalues (with the same multiplicities).

1. True

2. False

If x is an eigen value corresponding to eigen value of V of a matrix A . If a is any constant, then $x - a$ is an eigen value corresponding to eigen vector V is an of the matrix $A - aI$.

1. True

2. False

a ko agar aap lambda se replace kar dain tou baat clear ho jaye gi....labda ki jagah a use keea gaya hai tou is liay yeh working thora sa confuse karti hai...

Central difference method seems to be giving a better approximation, however it requires more computations.

1. True

2. False

Iterative algorithms can be more rapid than direct methods.

1. True (main nay true hi keea tha, aap isay dekh lena)

2. False

Central Difference method is the finite difference method.

1. True

2. False

MTH603 Numerical Analysis

Mid Term Examination - Spring 2006

Time Allowed: 90 Minutes

Question No. 1**Marks :**

Use bisection method to find the solution for
(Perform only three iterations.)

Question No. 2

$$2 + \cos(e^x - 2) - e^x = 0$$

on interval [0.5, 1.5]

Marks : 2

Bisection and false position methods are also known as bracketing method and are always

- Ⓐ Divergent
- Ⓑ Convergent

Question No. 3**Marks :**

Use Gauss Elimination method to solve the following system.

$$4x_1 - x_2 + x_3 = 8$$

$$2x_1 + 5x_2 + 2x_3 = 3$$

$$x_1 + 2x_2 + 4x_3 = 11$$

Question No. 4**Marks :**

The Inverse of a matrix can only be found if the matrix is

- Ⓐ Singular
- Ⓑ Non singular
- Ⓒ Scalar

☐ Diagonal

Question No. 5

Marks :

If $f(x)$ contains trigonometric, exponential or logarithmic functions then this equation is known as

- ☐ Transcendental equation
- ☐ Algebraic
- ☐ Polynomial
- ☐ Linear

Question No. 6

Marks :

In interpolation δ is used to represent the

- ☐ Forward difference
- ☐ Central difference
- ☐ Backward difference

Question No. 7

Marks :

The base of the decimal system is _____

- ☐ 10
- ☐ 0
- ☐ 2
- ☐ 8
- ☐ None of the above.

Question No. 8

Use Newton's Raphson Method to find the solution for (Perform only three iterations.)

Question No. 9

Marks : 10

$$x^3 + 3x^2 - 1 = 0 \text{ on } [-3, -2].$$

Marks : 10

Approximate $f(0.05)$ by using any of the interpolation technique.

x	0.0	0.2	0.4	0.6	0.8
F(x)	1.000	1.22140	1.49182	1.82212	2.22554

$$[(f_0 + f_n) + 2 \sum_{i=1}^{n-1} f_i]$$

is known as

► Simpson's 1/3 rd Rule

► Simpson's 3/8 rule

► Trapezoidal rule

Question No: 2 (Marks: 2) - Please choose one

Bisection method is method

► Open Method

► Bracketing Method

Question No: 3 (Marks: 2) - Please choose one

Which method is not used to solve problems related to integration?

► Runge-Kutta Method

► Simpson's 1/3rd rule

► Trapezoidal rule.

Question No: 7 (Marks: 10)

Interpolate the value of 0.25 using Newton's forward difference formula.

x	0.2	0.3	0.4	0.5	0.6
F(x)	0.2304	0.2788	0.3222	0.3617	0.3979

(Perform all the necessary calculation missing calculation and steps may deduct marks.)

Exact solution of $2/3$ is not exists.

TRUE

FALSE

The Jacobi's method is
a method of solving a matrix equation on a matrix that has ____ zeros along its main diagonal.

no

atleast one

A 3 x 3 identity matrix have three and _____ eigen values.

same
different

Eigenvalues of a symmetric matrix are all _____ .

real
complex
zero
positive

The Jacobi iteration converges, if A is strictly diagonally dominant.

TRUE
FALSE

Below are all the finite difference methods EXCEPT _____.

jacobi's method
newton's backward difference method
Stirling formula
Forward difference method

If n x n matrices A and B are similar, then they have the same eigenvalues (with the same multiplicities).

TRUE
FALSE

If A is a nxn triangular matrix (upper triangular, lower triangular) or diagonal matrix , the eigenvalues of A are the diagonal entries of A.

TRUE
FALSE

The characteristics polynomial of a 3x 3 identity matrix is _____, if x is the eigen values of the given 3 x 3 identity matrix. where symbol ^ shows power.

$(x-1)^3$
 $(x+1)^3$
 x^3-1
 x^3+1

Two matrices with the same characteristic polynomial need not be similar.

TRUE

FALSE

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i have attempt mth 603 today..... my paper was power method 5 marks , from vector 3 marks, Newton_Repshon's forward distance formula 5 marks.,define extrapolation 2 marks ,, and most of MCQ's from , Newton_Repshon's method ,or 18 to 22 lec's

question 29: Distinguish between Related and unrelated diversification with example 5marks

question 30 Five porters Model 5 marks

question 31: concentric diversification two example 3marks

question 32 significant of R&D for an organization 3marks

Which of the following period strategic management was considered to be cure for all problems?

Mid 1950s to mid 1960s

Mid 1960s to mid 1970s

Mid 1970s to mid 1980s

Mid 1980s to mid 1990s

Which of the following is not a pitfall an organization should avoid in strategic planning?

Select correct option:

Failing to involve key employees in all phases of planning

Involving all managers rather than delegating planning to a planner

Top managers not actively supporting the strategic planning process

Doing strategic planning only to satisfy accreditation or regulatory requirements

which of the following are the factors that concern the nature and direction of the economy in which a firm operates?

Select correct option:

Technological

Ecological

Social

Economic

Which of the following best describes this statement; “a Systematic and ethical process for gathering and analyzing information about the competition’s activities and general business trends to further a business’ own goals”?

Select correct option:

External assessment

Industry analysis

Competitive intelligence program

Business ethics

According to Porter, which strategy offers products or services to a small range of customers at the lowest price available on the market?

Select correct option:

Low cost

Best value

Cost focus

Differentiation

Long-term objectives includes all of the following EXCEPT:

Measurable

Reasonable

Varying

Consistent

Which one of the following is NOT is a basic mission of a competitive intelligence program?

To provide a general understanding of an industry

To provide a general understanding of a company’s competitors

To identify industry executives who could be hired by the firm

To identify potential moves a competitor might make that would endanger a firm

While preparing an External Factor Evaluation Matrix, a total score of 0.8 indicates that:

Firm is taking advantages of strengths and avoiding threats

Firm is taking no advantage of opportunities and is avoiding threats

Firm is not taking advantages of opportunities and is not avoiding threats

Firm is taking advantage of opportunities and is avoiding the threats

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Use the Regula-Falsi method to compute a real root of the equation

$$x^3 - 9x + 1 = 0, \text{ (Perform two iterations for each part)}$$

(i) if the root lies between 2 and 4

(ii) if the root lies between 2 and 3.

Solve the following system of equations by Crout's method

$$x_1 + 2x_2 + x_3 = 2$$

$$2x_1 + x_2 - 10x_3 = 0.182$$

$$2x_1 + 3x_2 - x_3 = 2$$

Find the numerical solution of

$$55x - 25z = -200$$

$$-37y - 4z = -250$$

$$-25x - 4y + 29z = 100$$

Using Gauss-Siedel Method with $(1, 1, 1)$ as the initial value. Perform two iterations only.

Solve $x^3 - 9x + 1 = 0$ for the root between $x = 2$ and $x = 4$ by the bisection method.
(Perform four iterations only)

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Question # 1 of 8 (**Start time: 08:34:31 PM**)

Total Marks: 1

Adams – Bashforth is a multistep method.

 **Select correct option:**

- ☐ TRUE
- ☐ FALSE

Click here to Save Answer & Move to Next Question

Total Marks: 1

Question # 2 of 8 (Start time: 08:35:01 PM)

Generally, Adams methods are superior if output at _____ points is needed.

▶ Select correct option:

- ☐ many
- ☐ two
- ☐ single
- ☐ at most



Click here to Save Answer & Move to Next Question

Total Marks: 1

Question # 3 of 8 (Start
time: 08:36:02 PM)

Total Marks: 1

In Trapezoidal rule, the integral is computed on each of the sub-intervals by using linear interpolating formula, i.e. for $n = 1$ and then summing them up to obtain the desired integral.

▶ Select correct option:

☐

TRUE

☐

FALSE



Click here to Save Answer & Move to Next Question

Question # 4 of 8 (Start time: 08:36:52 PM)

Euler's Method numerically computes the approximate _____ of a function.

▶ Select correct option:

☐

antiderivative

☐

derivative

☐

value

☐

error



Click here to Save Answer & Move to Next Question

Total Marks: 1

Question # 5 of 8 (Start time: 08:37:25 PM)

Multistep method does not improves the accuracy of the answer at each step.

▶ Select correct option:

- ☐ FALSE
- ☐ TRUE

▶ Click here to Save Answer & Move to Next Question

Total Marks: 1

Question # 6 of 8 (Start time: 08:37:56 PM)

The trapezoidal rule is a numerical method that approximates the value of a _____.

▶ Select correct option:

- ☐ indefinite integral
- ☐ definite integral
- ☐ improper integral
- ☐ function

▶ Click here to Save Answer & Move to Next Question

Total Marks: 1

Question # 7 of 8 (Start time: 08:38:47 PM)

Simpson's rule is a numerical method that approximates the value of a definite integral by using _____ polynomials.

▶ Select correct option:

- ☐ quadratic
- ☐ linear
- ☐ cubic
- ☐ quartic

▶ [Click here to Save Answer & Move to Next Question](#)

Total Marks: 1

Question # 8 of 8 (Start time: 08:39:23 PM)

In Simpson's Rule, we use parabolas to approximate each part of the curve. This proves to be very efficient as compared to Trapezoidal rule.

▶ Select correct option:

- ☐ TRUE
- ☐ FALSE

Question # 3 of 8 (Start time: 09:46:11 PM)

Total Marks: 1

The first langrange polynomial with equally spaced nodes produced the formula for _____.

▶ Select correct option:

- ☐ Simpson's rule
- ☐ Trapezoidal rule
- ☐ New ton's method
- ☐ Richardson's method

▶ Click here to Save Answer & Move to Next Question

/w EPDw UKMTY2

/w EWBgLgsdHjB

MC090406505 : Sumera Naz

Time Left 12 sec(s)

Quiz Start Time: 09:44 PM

Question # 4 of 8 (Start time: 09:46:48 PM)

Total Marks: 1

The need of numerical integration arises for evaluating the indefinite integral of a function that has no explicit antiderivative or whose antiderivative is not easy to obtain.

▶ Select correct option:

- ☐ TRUE
- ☐ FALSE

 Click here to Save Answer & Move to Next Question

/w EPDw UKMTY2

/w EWCgKl9pWZt

MC090406505 : Sumera Naz

Time
Left 73
sec(s)
)

Quiz Start Time: 09:44 PM

Question # 5 of 8 (Start time: 09:48:13 PM)

Total Marks: 1

The Trapezoidal Rule is an improvement over using rectangles because we have much less "missing" from our calculations. We used _____ to model the curve in trapezoidal Rule.

 Select correct option:

- ☐ straight lines
- ☐ curves
- ☐ parabolas
- ☐ constant

 Click here to Save Answer & Move to Next Question

/w EPDw UKMTY2

/w EWCgLo7tKA/

MC090406505 : Sumera Naz

Time
Left 69
sec(s)
)

Quiz Start Time: 09:44 PM

Question # 6 of 8 (Start time: 09:48:35 PM)

Total Marks: 1

The Euler method is numerically unstable because of _____ convergence of error.

 Select correct option:

- ☐ slow
- ☐ fast
- ☐ moderate
- ☐ no



Click here to Save Answer & Move to Next Question

/w EPDw UKMTY2

/w EWBgLyji72BA

MC090406505 : Sumera Naz

Time 74
Left sec(s)

Quiz Start Time: 09:44 PM

Question # 8 of 8 (Start time: 09:49:41 PM)

Total Marks: 1

Adams – Bashforth is a multistep method.

▶ Select correct option:

- ☐ TRUE
- ☐ FALSE



Click here to Save Answer & Move to Next Question

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/wEPDwUKMTY2M

/wEWBgK+3MqIA

Question # 1 of 8 (Start time: 01:24:49 PM)

Total Marks: 1

The need of numerical integration arises for evaluating the definite integral of a function that has no explicit _____ or whose antiderivative is not easy to obtain.

▶ Select correct option:

- ☒ antiderivative
- ☐ derivatives

/wEPDwUKMTY2M

/wEWBgLy4ZCpA

Question # 2 of 8 (Start time: 01:26:18 PM)

Total Marks: 1

In Runge – Kutta Method, we do not need to calculate higher order derivatives and find greater accuracy.

▶ Select correct option:

- ☐ TRUE
- ☒ FALSE

/wEPDwUKMTY2M

/wEWBgLlwO3cC

Question # 3 of 8 (Start time: 01:27:35 PM)

Total Marks: 1

An indefinite integral may _____ in the sense that the limit defining it may not exist.

▶ Select correct option:

- ☒ diverge

☐ converge

/wEPDwUKMTY2M /wEWCgKMu86bL

Question # 5 of 8 (Start time: 01:30:21 PM)

Total Marks: 1

The Trapezoidal Rule is an improvement over using rectangles because we have much less "missing" from our calculations. We used _____ to model the curve in trapezoidal Rule.

▶ Select correct option:

☒ straight lines

☐ curves

☐ parabolas

☐ constant

/wEPDwUKMTY2M /wEWBgK6oIaxB

Question # 6 of 8 (Start time: 01:31:18 PM)

Total Marks: 1

An improper integral is the limit of a definite integral as an endpoint of the interval of integration approaches either a specified real number or ∞ or $-\infty$ or, in some cases, as both endpoints approach limits.

▶ Select correct option:

☒ TRUE

☐ FALSE

 Click here to Save Answer & Move to Next Question

/wEPDwUKMTY2M

/wEWBgkPg8yyD

Question # 7 of 8 (Start time: 01:32:33 PM)

Total Marks: 1

Euler's Method numerically computes the approximate derivative of a function.

 Select correct option:

- ☒ FALSE
- ☐ TRUE

/wEPDwUKMTY2M

/wEWCgKK6eLRC

Question # 8 of 8 (Start time: 01:33:57 PM)

Total Marks: 1

If we wanted to find the value of a definite integral with an infinite limit, we can instead replace the infinite limit with a variable, and then take the limit as this variable goes to _____.

 Select correct option:

- ☐ constant
- ☐ finite
- ☒ infinity
- ☐ zero

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/wEPDwUKMTY2M

/wEWBgK+3MqIA

Question # 1 of 8 (Start time: 01:24:49 PM)

Total Marks: 1

The need of numerical integration arises for evaluating the definite integral of a function that has no explicit _____ or whose antiderivative is not easy to obtain.

▶ **Select correct option:**

☐ antiderivative

☐ derivatives



Click here to Save Answer & Move to Next Question

/wEPDwUKMTY2M

/wEWBgLy4ZCpA

Time
Left

59
sec(
s)

Question # 2 of 8 (Start time: 01:26:18 PM)

Total Marks: 1

In Runge – Kutta Method, we do not need to calculate higher order derivatives and find greater accuracy.

▶ **Select correct option:**

☐ TRUE

☐ FALSE



Click here to Save Answer & Move to Next Question

/wEPDwUKMTY2M

/wEWBgLlwO3cC

Time
Left

79
sec(
s)

Question # 3 of 8 (Start time: 01:27:35 PM)

Total Marks: 1

An indefinite integral may _____ in the sense that the limit defining it may not exist.

▶ Select correct option:

- ☐ diverge
- ☐ converge



Click here to Save Answer & Move to Next Question

/wEPDwUKMTY2M

/wEWCgKM86bC

Time
Left

85
sec(
s)

Question # 5 of 8 (Start time: 01:30:21 PM)

Total Marks: 1

The Trapezoidal Rule is an improvement over using rectangles because we have much less "missing" from our calculations. We used _____ to model the curve in trapezoidal Rule.

▶ Select correct option:

- ☐ straight lines
- ☐ curves
- ☐ parabolas
- ☐ constant



Click here to Save Answer & Move to Next Question

/wEPDwUKMTY2M

/wEWBgK6oIaxB

Time
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83
sec(
s)

Question # 6 of 8 (Start time: 01:31:18 PM)

Total Marks: 1

An improper integral is the limit of a definite integral as an endpoint of the interval of integration approaches either a specified real number or ∞ or $-\infty$ or, in some cases, as both endpoints approach limits.

▶ Select correct option:

- ☐ TRUE
- ☐ FALSE



Click here to Save Answer & Move to Next Question

/wEPDwUKMTY2M

/wEWBgKPg8yyD

Time
Left

84
sec(
s)

Question # 7 of 8 (Start time: 01:32:33 PM)

Total Marks: 1

Euler's Method numerically computes the approximate derivative of a function.

▶ Select correct option:

- ☐ FALSE
- ☐ TRUE



Click here to Save Answer & Move to Next Question

/wEPDwUKMTY2M

/wEWCgKK6eLRD

Time
Left

84
sec(
s)

Question # 8 of 8 (Start time: 01:33:57 PM)

Total Marks: 1

If we wanted to find the value of a definite integral with an infinite limit, we can instead replace the infinite limit with a variable, and then take the limit as this variable goes to _____.

▶ Select correct option:

- ☐ constant
- ☐ finite
- ☐ infinity
- ☐ zero



Click here to Save Answer & Move to Next Question

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MTH603-Numerical Analysis SEMESTER FALL 2005

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Approximate the integral $\int_1^{1.6} \frac{2x}{x^2+1} dx$ using Simpson's 1/3 rule and calculate the percentage error.
(Take result up to 4 decimal places)

Note: In order to get full marks do all necessary steps.

Construct a forward difference table for the following values

x	0.1	0.3	0.5	0.7	0.9	1.1	1.3
---	-----	-----	-----	-----	-----	-----	-----

y 0.003 0.067 0.148 0.248 0.37 0.518 0.697

Note : In order to get full marks do all necessary steps.

Solve the system

$$4x + 3y = 24$$

$$3x + 4y - z = 30$$

$$-y + 4z = -24$$

by Gauss Seidal Method, taking $(0, 0, 0)^t$ as initial approximation(Two iterations only and take result up to 4 decimal places)

Note : In order to get full marks do all necessary steps

Let $f(x) = x + \frac{2}{x}$, use cubic Lagrange interpolation based on the nodes $x_0 = 0.5, x_1 = 1, x_2 = 2$ and $x_3 = 2.5$ to approximate $f(1.5)$ and $f(1.3)$.

Note : In order to get full marks do all necessary steps**Solution**

Approximate the Dominant Eigenvalue and corresponding Eigenvector for the matrix

$$\begin{bmatrix} 0 & 11 & -5 \\ -2 & 17 & -7 \\ -4 & 26 & -10 \end{bmatrix}$$

by using Power Method. Start with $X_0 = (1, 1, 1)^t$. (Five iterations only and take result up to 4 decimal places)

Note : In order to get full marks do all necessary steps

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NUMERICAL ANALYSIS

Paper Final Term (Held: 25th Feb 2010)

No	Questions	Mks
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.																		
1	Find value of given data by Adam Moulton's method	10																
2	Find value of given data by Dividend Difference Composite method	10																
3	Draw backward difference tables for given Data <table><tr><td>x</td><td></td><td></td><td></td></tr><tr><td>y</td><td></td><td></td><td></td></tr></table> <table><tr><td>x</td><td></td><td></td><td></td></tr><tr><td>y</td><td></td><td></td><td></td></tr></table>	x				y				x				y				5+5
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y																		
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y																		
4	Write Simpson's 1/3 formula	2																
5	Find value by Euler's Method	3																
6	Find value of K1 by 2 nd Order R-K method	2																
7	Convergence is used when -----	1																
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9	Newton Raphson method is method • Bracketing Method • Open • Random • none	1																
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12	Find value of y'(3) from given table. <table><tr><td>x</td><td></td><td></td><td></td></tr><tr><td>y</td><td></td><td></td><td></td></tr></table>	x				y				2								
x																		
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13	Find value of y'(0.3) by Lagrange's Method <table><tr><td>x</td><td></td><td></td><td></td></tr><tr><td>y</td><td></td><td></td><td></td></tr></table>	x				y				3								
x																		
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14	For Simpson's 1/3 rule no.of intervals must be • 1 • 3 • 5 • 8	1																
15	For Simpson's 1/3 rule valid no.of intervals are • 1	1																

	• 3 • 5 • 8																										
16	For Simpson's 3/8 rule no.of intervals must be • 10 • 11 • 12 • 14 http://vustudents.ning.com/	1																									
17	Find the value of $y'(1)$ from given forward difference table <table><tr><th>x</th><th>y</th><th>Δy</th><th>Δy^2</th><th>Δy^3</th></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table>	x	y	Δy	Δy^2	Δy^3																					2
x	y	Δy	Δy^2	Δy^3																							

Numerical analysis mth603 paper

Numerical Analysis numerical paper 2009

The paper was very easy.

The mcq's were really easy.

Most of the mcq's were from the last 5 lecture.

and from jacobi's method and other's.

then the logical mcq's like

s inverse * s = I

it was repeated twice.

one 5 mark question was from newton's rapson method

one 10 mark question was from lecture 11 page 69 example

best of luck

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NUMERICAL ANALYSIS

Paper Final Term (Held: 25th Feb 2010)

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15	For Simpson's 1/3 rule valid no.of intervals are • 1 • 3 • 5	1																

	• 8																										
16	For Simpson's 3/8 rule no.of intervals must be • 10 • 11 • 12 • 14 http://vustudents.ning.com/	1																									
17	Find the value of $y'(1)$ from given forward difference table <table><tr><th>x</th><th>y</th><th>Δy</th><th>Δy^2</th><th>Δy^3</th></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr><tr><td></td><td></td><td></td><td></td><td></td></tr></table>	x	y	Δy	Δy^2	Δy^3																					2
x	y	Δy	Δy^2	Δy^3																							

Numerical analysis mth603 paper

Numerical Analysis numerical paper 2009

The paper was very easy.

The mcq's were really easy.

Most of the mcq's were from the last 5 lecture.

and from jacobi's method and other's.

then the logical mcq's like

s inverse * s = I

it was repeated twice.

one 5 mark question was from newton's rapson method

one 10 mark question was from lecture 11 page 69 example

best of luck

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FINAL TERM EXAMINATION
Spring 2010
MTH603- Numerical Analysis (Session - 2)

Ref No: 1508683
Time: 90 min
Marks: 60

Question No: 1 (Marks: 1) - Please choose one

Symbol used for forward differences is

▶ ∇

▶ Δ

▶ δ

▶ μ

Question No: 2 (Marks: 1) - Please choose one

The relationship between central difference operator and the shift operator is given by

▶ $\delta = E - E^{-1}$

▶ $\delta = E + E^{-1}$

▶ $\delta = E^{\frac{1}{2}} + E^{-\frac{1}{2}}$

▶ $\delta = E^{\frac{1}{2}} - E^{-\frac{1}{2}}$

Question No: 3 (Marks: 1) - Please choose one

Muller's method requires -----starting points

▶ 1

▶ 2

▶ 3

▶ 4

Question No: 4 (Marks: 1) - Please choose one

If S is an identity matrix, then

▶ $S^{-1} = S$

▶ $S^t = S$

▶ *All are true*

▶ $S^{-1} = S^t$

Question No: 5 (Marks: 1) - Please choose one

_____ If we retain $r+1$ terms in Newton's forward difference formula, we obtain a polynomial of degree ---- agreeing with y_x at $x_0, x_1, \dots, x_r$

- ▶ $r+2$
- ▶ $r+1$
- ▶ r
- ▶ $r-1$

Question No: 6 (Marks: 1) - Please choose one

_____ P in Newton's forward difference formula is defined as

- ▶ $p = \left(\frac{x - x_0}{h} \right)$
- ▶ $p = \left(\frac{x + x_0}{h} \right)$
- ▶ $p = \left(\frac{x + x_n}{h} \right)$
- ▶ $p = \left(\frac{x - x_n}{h} \right)$

Question No: 7 (Marks: 1) - Please choose one

_____ Octal number system has the base -----

- ▶ 2
- ▶ 8
- ▶ 10
- ▶ 16

Question No: 8 (Marks: 1) - Please choose one

_____ Newton's divided difference interpolation formula is used when the values of the independent variable are

- ▶ Equally spaced
- ▶ Not equally spaced
- ▶ Constant
- ▶ None of the above

Question No: 9 (Marks: 1) - Please choose one

Given the following data

x	0	1	2	4
$f(x)$	1	1	2	5

Value of $f(2,4)$ is

- ▶ 1.5
- ▶ 3
- ▶ 2
- ▶ 1

Question No: 10 (Marks: 1) - Please choose one

If $y(x)$ is approximated by a polynomial $P_n(x)$ of degree n then the error is given by

- ▶ $\varepsilon(x) = y(x) + P_n(x)$
- ▶ $\varepsilon(x) = y(x) - P_n(x)$
- ▶ $\varepsilon(x) = P_n(x) - y(x)$
- ▶ $\varepsilon(x) = y(x) \times P_n(x)$

Question No: 11 (Marks: 1) - Please choose one

Let I denotes the closed interval spanned by $x_0, x_1, x_2, x_3, x_4, x_5, x_6, x_7, \bar{x}$. Then $F(x)$ vanishes -----times in the interval I .

- ▶ $n-1$
- ▶ $n+2$
- ▶ n
- ▶ $n+1$

Question No: 12 (Marks: 1) - Please choose one

Differential operator in terms of forward difference operator is given by

- ▶ $D = \frac{1}{h} \left(\Delta + \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} + \frac{\Delta^4}{4!} + \frac{\Delta^5}{5!} + \dots \right)$

► $D = \frac{1}{h} \left(\Delta + \frac{\Delta^2}{2} + \frac{\Delta^3}{3} + \frac{\Delta^4}{4} + \frac{\Delta^5}{5} + \dots \right)$

► $D = \frac{1}{h} \left(\Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \frac{\Delta^5}{5} - \dots \right)$

► $D = \frac{1}{h} \left(\Delta - \frac{\Delta^2}{2!} + \frac{\Delta^3}{3!} - \frac{\Delta^4}{4!} + \frac{\Delta^5}{5!} - \dots \right)$

Question No: 13 (Marks: 1) - Please choose one

Finding the first derivative of $f(x)$ at $x=0.4$ from the following table:

x	0.1	0.2	0.3	0.4
$f(x)$	1.10517	1.22140	1.34986	1.49182

Differential operator in terms of -----will be used.

- Forward difference operator
- Backward difference operator
- Central difference operator
- None of the given choices

Question No: 14 (Marks: 1) - Please choose one

For the given table of values

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

$f'(0.1)$, using two-point equation will be calculated as.....

- -0.5
- 0.5
- 0.75
- -0.75

Question No: 15 (Marks: 1) - Please choose one

Simpson's 1/3 rule, $f(x)$ is of the form

In

- ▶ $ax + b$
- ▶ $ax^2 + bx + c$
- ▶ $ax^3 + bx^2 + cx + d$
- ▶ $ax^4 + bx^3 + cx^2 + dx + e$

Question No: 16 (Marks: 1) - Please choose one

While integrating $I = \int_a^b f(x)dx$, h , width of the interval, is found by the formula-----.

- ▶ $\frac{b-a}{n}$
- ▶ $\frac{b+a}{n}$
- ▶ $\frac{a-b}{n}$
- ▶ None of the given choices

Question No: 17 (Marks: 1) - Please choose one

To apply Simpson's 1/3 rule, valid number of intervals are.....

- ▶ 7
- ▶ 8
- ▶ 5
- ▶ 3

Question No: 18 (Marks: 1) - Please choose one

For the given table of values

x	0.2	0.3	0.4	0.5	0.6	0.7
$f(x)$	0.425	0.475	0.400	0.452	0.525	0.575

$f''(0.2)$, using three-point equation will be calculated as

- ▶ 17.5
- ▶ 12.5
- ▶ 7.5
- ▶ -12.5

Question No: 19 (Marks: 1) - Please choose one

_____ To
apply Simpson's 1/3 rule, the number of intervals in the following must be

- ▶ 2
- ▶ 3
- ▶ 5
- ▶ 7

Question No: 20 (Marks: 1) - Please choose one

_____ To
apply Simpson's 3/8 rule, the number of intervals in the following must be

- ▶ 10
- ▶ 11
- ▶ 12
- ▶ 13

Question No: 21 (Marks: 1) - Please choose one

_____ If
the root of the given equation lies between a and b, then the first approximation to the root of the equation by bisection method is

- ▶ $\frac{(a+b)}{2}$
- ▶ $\frac{(a-b)}{2}$
- ▶ $\frac{(b-a)}{2}$
- ▶ None of the given choices

Question No: 22 (Marks: 1) - Please choose one

_____lies in the category of iterative method.

- ▶ Bisection Method
- ▶ Regula Falsi Method
- ▶ Secant Method
- ▶ None of the given choices

Question No: 23 (Marks: 1) - Please choose one

_____ For
the equation $x^3 + 3x - 1 = 0$, the root of the equation lies in the interval.....

- ▶ (1, 3)
- ▶ (1, 2)
- ▶ (0, 1)
- ▶ (1, 2)

Question No: 24 (Marks: 1) - Please choose one

Rate of change of any quantity with respect to another can be modeled by

- ▶ An ordinary differential equation
- ▶ A partial differential equation
- ▶ A polynomial equation
- ▶ None of the given choices

Question No: 25 (Marks: 1) - Please choose one

If $\frac{dy}{dx} = f(x, y)$

Then the integral of this equation is a curve in

- ▶ None of the given choices
- ▶ xt-plane
- ▶ yt-plane
- ▶ xy-plane

Question No: 26 (Marks: 1) - Please choose one

In solving the differential equation

$$y' = x + y ; y(0.1) = 1.1$$

$h = 0.1$, By Euler's method $y(0.2)$ is calculated as

- ▶ 1.44
- ▶ 1.11
- ▶ 1.22
- ▶ 1.33

Question No: 27 (Marks: 1) - Please choose one

In second order Runge-Kutta method

k_1 is given by

- ▶ $k_1 = hf(x_n, y_n)$
- ▶ $k_1 = 2hf(x_n, y_n)$
- ▶ $k_1 = 3hf(x_n, y_n)$
- ▶ None of the given choices

Question No: 28 (Marks: 1) - Please choose one

fourth order Runge-Kutta method, k_2 is given by

- ▶ $k_2 = hf(x_n + \frac{h}{2}, y_n + \frac{k_1}{2})$
- ▶ $k_2 = hf(x_n + \frac{h}{3}, y_n + \frac{k_1}{3})$
- ▶ $k_2 = hf(x_n - \frac{h}{3}, y_n - \frac{k_1}{3})$
- ▶ $k_2 = hf(x_n - \frac{h}{2}, y_n - \frac{k_1}{2})$

Question No: 29 (Marks: 1) - Please choose one

fourth order Runge-Kutta method, k_4 is given by

- ▶ $k_3 = hf(x_n + 2h, y_n + 2k_3)$
- ▶ $k_3 = hf(x_n - h, y_n - k_3)$
- ▶ $k_3 = hf(x_n + h, y_n + k_3)$
- ▶ None of the given choices

Question No: 30 (Marks: 1) - Please choose one

Adam-Moulton P-C method is derived by employing

- ▶ Newton's backward difference interpolation formula
- ▶ Newton's forward difference interpolation formula
- ▶ Newton's divided difference interpolation formula
- ▶ None of the given choices

Question No: 31 (Marks: 2)

If $F(h) = 256.2354$ and $F(\frac{h}{2}) = 257.1379$, then find $F_1(\frac{h}{2})$ using Richardson's extrapolation limit.

Question No: 32 (Marks: 2)

Evaluate the integral

$$\int_0^{\frac{\pi}{2}} (\cos x + 2) dx$$

Using Simpson's 3/8 rule

Take $h = \frac{\pi}{4}$

Question No: 33 (Marks: 2)

Write a general formula for Modified Euler's method of solving the given differential equation.

Question No: 34 (Marks: 3)

Evaluate the integral

$$\int_0^4 x^2 dx$$

Using Trapezoidal rule

Take $h=1$

Question No: 35 (Marks: 3)

Evaluate the integral

$$\int_3^5 (\log x + 2) dx$$

Using Simpson's 3/8 rule

Take $h=1$

Question No: 36 (Marks: 3)

Write a formula for finding the value of k_3 in Fourth-order R-K method.

Question No: 37 (Marks: 5)

Find Newton's forward difference table from the following data.

x	0.0	0.1	0.2	0.3	0.4
$f(x)$	1	0.9048	0.8187	0.7408	0.6703

Question No: 38 (Marks: 5)

Evaluate the integral

$$\int_0^3 (x^2 + x) dx$$

Using Simpson's 3/8 rule

Take $h=1$

Question No: 39 (Marks: 5)

Use

Runge-Kutta Method of order four to find the values of k_1, k_2, k_3 and k_4 for the initial value problem

$$y' = \frac{1}{2}(2x^3 + y), y(1) = 2 \text{ taking } h = 0.1$$

(THE END)